

Fishing Harbor Weather Alert App

114-2 APP Development Final Project

Topic: Maritime and Ocean Application

Android app for harbor weather, wave, tide, safety alert, and map lookup.

Android Studio Room Retrofit Google Maps Flask

2. Project Background

- Taiwan has many harbors used for travel, fishing, tourism, and daily activities.
- Wind speed, wave height, rain probability, and tide status can affect coastal safety.

3. User Pain Points

- Weather and marine information is scattered across different services.
- Users need a fast way to judge whether harbor conditions are safe.
- Frequent harbor visitors need favorites and quick re-entry.

4. Core Features

- Dashboard with harbor cards and safety level badges.
- Detail page with weather, temperature, rain, wind, wave, tide, and update time.
- Favorites, alert settings, Google Maps marker, and GPS position.

5. Android Architecture

- Java + XML Layout + Material Components.
- RecyclerView for dynamic harbor list and Activity navigation.
- Room stores harbors, favorites, and alert rules; SharedPreferences stores user preferences.

6. API and Data

- Retrofit structure is prepared for external JSON weather API.
- When API keys are empty, WeatherRepository returns sample weather data for classroom demo.
- SafetyEvaluator maps wind/wave values to SAFE, CAUTION, and DANGER.

7. Maps and Backend

- MapActivity uses Google Maps Fragment and location permission handling.
- Flask backend provides /health, /harbors, POST /alert-log, and GET /alert-log.
- The backend can be deployed to AWS EC2 for bonus demonstration.

8. Result and Next Steps

- The project builds and runs in Android Studio with Gradle Wrapper.
- Future work: real Central Weather Administration marine API, background alerts, and notification push.